

Program 5

5. Consider the following database of student enrolment in courses and books adopted each course.

STUDENT (regno: string, name: string, major: string, bdate: date)

COURSE (course: int, cname: string, dept: string)

ENROLL (#regno: string, course#: int, sem:int marks: int)

TEXT (book-ISBN: int, book-title: string, publisher: string, author: string)

BOOK_ADOPTION (course#: int, sem: int, book-ISBN#: int)

Create the above tables by properly specifying the primary keys and the foreign Keys.

- i. Enter at least five tuples for each relation.
- ii. Demonstrate how you add a textbook to the database and make this book be adapted by some department.
- iii. Produce list of textbooks (include Course#, Book-ISBN, Book-title) in the alphabetical order for courses offered by the CS department that use more than two books.
- iv. List any department that has its adopted books published by a specific publisher.

Create the above tables by properly specifying the primary keys and the foreign Keys.

```
mysql> create database labprog5;  
Query OK, 1 row affected (0.05 sec)
```

```
mysql> use labprog5;  
Database changed
```

```
mysql> create table student (regno varchar(20),name varchar(20), major varchar(20),  
-> bdate date, primary key(regno));  
Query OK, 0 rows affected (0.10 sec)
```

```
mysql> create table course (courseno int , cname varchar(20),  
-> department varchar(20), primary key(courseno));  
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> create table enroll (regno varchar(20), courseno int,  
-> sem int, marks int,  
-> foreign key(regno) references student (regno),  
-> foreign key(courseno) references course(courseno));  
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> create table text (book_isbn int, book_title varchar(20),  
-> publisher varchar(20), author varchar(20),  
-> primary key(book_isbn));  
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> create table book_adoption(courseno int,
-> sem int, book_isbn int,
-> foreign key(courseno) references course(courseno),
-> foreign key(book_isbn) references text(book_isbn));
Query OK, 0 rows affected (0.06 sec)
```

i. Enter at least five tuples for each relation.

```
mysql> insert into student values('ab11','chirag','physics','1988-03-17');
Query OK, 1 row affected (0.03 sec)
```

```
mysql> insert into student values ('cd22','veeresh','maths','2000-12-18');
Query OK, 1 row affected (0.02 sec)
```

```
mysql> insert into student values ('ef33','banty','economics','1986-09-19');
Query OK, 1 row affected (0.03 sec)
```

```
mysql> insert into student values ('gh44','rani','computers','1989-11-20');
Query OK, 1 row affected (0.05 sec)
```

```
mysql> insert into student values ('ij55', 'sonu','electronics','1989-07-21');
Query OK, 1 row affected (0.05 sec)
```

```
mysql> select * from student;
+-----+-----+-----+-----+
| regno | name   | major   | bdate   |
+-----+-----+-----+-----+
| ab11  | chirag | physics | 1988-03-17 |
| cd22  | veeresh | maths   | 2000-12-18 |
| ef33  | banty  | economics | 1986-09-19 |
| gh44  | rani   | computers | 1989-11-20 |
| ij55  | sonu   | electronics | 1989-07-21 |
+-----+-----+-----+-----+
5 rows in set (0.05 sec)
```

```
mysql> insert into course values (111,'bca','cs');
Query OK, 1 row affected (0.03 sec)
```

```
mysql> insert into course values (222,'bba','mgmt');
Query OK, 1 row affected (0.01 sec)
```

```
mysql> insert into course values (333,'bcom','commerce');
Query OK, 1 row affected (0.03 sec)
```

```
mysql> insert into course values (444,'pgdca','cs');
Query OK, 1 row affected (0.02 sec)
```

```
mysql> insert into course values (555,'bsc','cs');
Query OK, 1 row affected (0.05 sec)
```

```
mysql> select * from course;
+-----+-----+-----+
| courseno | cname | department |
+-----+-----+-----+
| 111 | bca | cs |
| 222 | bba | mgmt |
| 333 | bcom | commerce |
| 444 | pgdca | cs |
| 555 | bsc | cs |
+-----+-----+-----+
5 rows in set (0.01 sec)
```

```
mysql> insert into enroll values ('ab11', 111, 1001, 750);
Query OK, 1 row affected (0.03 sec)
```

```
mysql> insert into enroll values ('cd22', 222, 1002, 900);
Query OK, 1 row affected (0.03 sec)
```

```
mysql> insert into enroll values ('ef33', 333, 1003, 870);
Query OK, 1 row affected (0.03 sec)
```

```
mysql> insert into enroll values ('gh44', 444, 1004, 920);
Query OK, 1 row affected (0.05 sec)
```

```
mysql> insert into enroll values ('ij55', 555, 1005, 980);
Query OK, 1 row affected (0.03 sec)
```

```
mysql> select * from enroll;
+-----+-----+-----+-----+
| regno | courseno | sem | marks |
+-----+-----+-----+-----+
| ab11 | 111 | 1001 | 750 |
| cd22 | 222 | 1002 | 900 |
| ef33 | 333 | 1003 | 870 |
| gh44 | 444 | 1004 | 920 |
| ij55 | 555 | 1005 | 980 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> insert into text values (1111,'c-programming', 'tata', 'balaguruswamy');
Query OK, 1 row affected (0.05 sec)
```

```
mysql> insert into text values (2222,'economics','birla','rohan');
Query OK, 1 row affected (0.02 sec)
```

```
mysql> insert into text values (3333,'dbms','romeo','nawate');
Query OK, 1 row affected (0.05 sec)
```

```
mysql> insert into text values (4444,'commerce','juliet','b-mahesh');
Query OK, 1 row affected (0.03 sec)
```

```
mysql> insert into text values (5555,'Physics','amar','kavitha');
Query OK, 1 row affected (0.05 sec)
```

```
mysql> select * from text;
```

book_isbn	book_title	publisher	author
1111	c-programming	tata	balaguruswamy
2222	economics	birla	rohan
3333	dbms	romeo	nawate
4444	commerce	juliet	b-mahesh
5555	Physics	amar	kavitha

5 rows in set (0.00 sec)

```
mysql> delete from text;
```

Query OK, 5 rows affected (0.05 sec)

```
mysql> select * from text;
```

Empty set (0.00 sec)

```
mysql> insert into text values (1000,'c-programming', 'tata', 'balaguruswamy');
```

Query OK, 1 row affected (0.03 sec)

```
mysql> insert into text values (2000,'economics','birla','rohan');
```

Query OK, 1 row affected (0.03 sec)

```
mysql> insert into text values (3000,'dbms','romeo','nawate');
```

Query OK, 1 row affected (0.03 sec)

```
mysql> insert into text values (4000,'commerce','juliet','b-mahesh');
```

Query OK, 1 row affected (0.12 sec)

```
mysql> insert into text values (5000,'Physics','amar','kavitha');
```

Query OK, 1 row affected (0.03 sec)

```
mysql> select * from text;
```

book_isbn	book_title	publisher	author
1000	c-programming	tata	balaguruswamy
2000	economics	birla	rohan
3000	dbms	romeo	nawate
4000	commerce	juliet	b-mahesh
5000	Physics	amar	kavitha

5 rows in set (0.00 sec)

```
mysql> insert into book_adoption values (111, 1001, 1000);
```

Query OK, 1 row affected (0.03 sec)

```
mysql> insert into book_adoption values (222, 1002, 2000);
```

Query OK, 1 row affected (0.02 sec)

```
mysql> insert into book_adoption values (333, 1003, 3000);
```

Query OK, 1 row affected (0.01 sec)

```
mysql> insert into book_adoption values (444, 1004, 4000);
Query OK, 1 row affected (0.03 sec)
```

```
mysql> insert into book_adoption values (555, 1005, 5000);
Query OK, 1 row affected (0.03 sec)
```

```
mysql> select * from book_adoption;
```

courseno	sem	book_isbn
111	1001	1000
222	1002	2000
333	1003	3000
444	1004	4000
555	1005	5000

5 rows in set (0.00 sec)

ii) Demonstrate how you add a new text book to the database and make this book to be adopted by some department.

```
mysql> insert into text values (32, 'java','pearson','somerville');
Query OK, 1 row affected (0.03 sec)
```

```
mysql> insert into book_adoption values (111, 1001, 32);
Query OK, 1 row affected (0.05 sec)
```

```
mysql> select * from text;
```

book_isbn	book_title	publisher	author
32	java	pearson	somerville
1000	c-programming	tata	balaguruswamy
2000	economics	birla	rohan
3000	dbms	romeo	nawate
4000	commerce	juliet	b-mahesh
5000	Physics	amar	kavitha

6 rows in set (0.00 sec)

```
mysql> select * from book_adoption;
```

courseno	sem	book_isbn
111	1001	1000
222	1002	2000
333	1003	3000
444	1004	4000
555	1005	5000
111	1001	32

6 rows in set (0.00 sec)

iii) Produce a list of text books(include course#, Book-isbn, Book-title) in the alphabetical order for course offered by the 'cs' department that use more than two books .

```
mysql> select course.courseno, text.book_isbn, text.book_title
-> from course, text, book_adoption
-> where course.courseno=book_adoption.courseno
-> and book_adoption.book_isbn=text.book_isbn
-> and course.courseno in (select course.courseno from book_adoption, course
-> where course.courseno=book_adoption.courseno and course.department like 'cs'
-> group by course.courseno having count(book_adoption.book_isbn)>1)
-> order by course.courseno;
```

courseno	book_isbn	book_title
111	1000	c-programming
111	32	java

2 rows in set (0.03 sec)

iv) List any department that has all its adopted books published by a specific publisher.

```
mysql> select distinct course.department from course ,book_adoption, text
-> where book_adoption.book_isbn=text.book_isbn
-> and publisher like 'tata'
-> and Course.courseno=book_adoption.courseno;
```

department
cs

1 row in set (0.00 sec)

```
mysql> commit;
Query OK, 0 rows affected (0.00 sec)
```